

# HealthFirst Digital Dental Camp Platform

## Technical Specification Document

### 1. UI Wireframes (Concept Layouts)

```
Registration Page
-----
[ Logo ]
Dental Screening Registration

Name: _____
Age:  _____
Gender: _____
Phone: _____
Organization: _____

[ Register ]


Doctor Dashboard
-----
Total Registered: ____
Screened:  ____
Pending:  ____

Patient Queue
-----
ID | Name | Status | Start Exam
-----

Screening Form
-----
Oral Hygiene:  Good / Fair / Poor
Caries:        Yes / No
Gingivitis:    Yes / No
Malocclusion:  Yes / No
Notes: _____

Treatment Advice
Scaling / Filling / Extraction / RCT

[ Generate Report ]
```

### 2. Database Schema

| Table      | Fields                                                                          |
|------------|---------------------------------------------------------------------------------|
| Patients   | patient_id, name, age, gender, phone, organization, created_at                  |
| Screenings | screening_id, patient_id, oral_hygiene, caries, gingivitis, malocclusion, notes |
| Treatments | treatment_id, patient_id, scaling, filling, extraction, rct                     |
| Reports    | report_id, patient_id, report_file, created_at                                  |
| Users      | user_id, name, role, login_email, password_hash                                 |

### 3. API Structure

- POST /api/register – register participant
- GET /api/patients – fetch patient list
- GET /api/patient/{id} – fetch patient details
- POST /api/screening – submit screening findings
- POST /api/report – generate dental report
- GET /api/report/{patient\_id} – download patient report
- GET /api/analytics – camp analytics data

## 4. User Flow Diagrams

Participant Flow  
-----  
Scan Registration Link  
↓  
Fill Registration Form  
↓  
Patient ID Generated  
↓  
Added to Screening Queue

Doctor Flow  
-----  
Login  
↓  
Open Patient Queue  
↓  
Select Patient  
↓  
Perform Screening  
↓  
Submit Findings  
↓  
Generate Dental Report

Admin Flow  
-----  
Login  
↓  
View Camp Dashboard  
↓  
Analyze Data  
↓  
Export Reports

## 5. Future Enhancements

- AI report summarization
- Automated WhatsApp report delivery
- AI dental risk scoring
- Intraoral photo analysis